Nesterov's Accelerated Gradient Descent's Speed

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Introduction

Gradient Descent, originally discussed by Cauchy et al. 1847, is the simplest strategy to minimize a differentiable function f on $\mathbb{R}_n$. It is well-known that the convergence rate for this algorithm is $O(1/t)$. Here, we discuss the lower bounds on the convergence rates for β-smooth convex functions in $O(1/t^2)$, and emphasize how Nesterov's Accelerated Gradient Descent achieves this bound. The main source of reference for this poster is from Chapter 3 of Bubeck 2015.

Notation

$\delta f(x)$ is the set of **subgradients** of the point $x \in \mathbb{R}^n$. Then the subgradient $g \in \delta f(x)$ satisfies $f(x) - f(y) \leq g^\top(x-y)$.
$e_1, \dots, e_n$ is the canonical basis of $\mathbb{R}^n$.
$x^* \in \mathbb{R}^n$ is a point such that it is the minima in any optimisation problem, i.e. $f(x^*) = \min_{x \in \mathbb{R}^n} f(x)$. We always assume x^* exists.
$\|\cdot\|$ is always the Euclidean norm $\|x\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$ for $x \in \mathbb{R}^n$

Fastest theoretical speed for Gradient Descent

Theorem (1) states that, no matter the gradient method provided, there always exists a function f such that when the gradient method is applied on minimizing such f, the convergence rate is lower bounded as $O(1/t^2)$.

Theorem (1)

Let $t \leq \frac{n-1}{2}$ and $\beta > 0$. There exists a β-smooth convex function f such that for any black-box procedure satisfying $x_{t+1} \in Span(g_1, \dots, g_t)$ with $g_s \in \delta f(x_s)$:

$$\min_{1 \leq s \leq t} f(x_s) - f(x^*) \geq \frac{3\beta \|x_1 - x^*\|^2}{32 \quad (t+1)^2}$$

We observe that Theorem 3 attains this complexity bound of $O(1/t^2)$

Crafting a β-smooth function f

To construct a suitable function f that satisfies (subsequently, proves) Theorem (1), Bubeck 2015 constructs a symmetric and tridiagonal matrix

$$(A_k)_{i,j} = \begin{cases} 2, & i = j\ ,\ i \leq k \\ -1, & j \in \{i-1, i+1\},\ i \leq k,\ j \neq k+1 \\ 0, & \text{otherwise} \end{cases}$$

and defines f as:

$$f_s(x) = \frac{\beta}{8} x^\top A_s x - \frac{\beta}{4} x^\top e_1$$

For $s \leq t$, we have $x_s(i) = 0$ for $i = s, \dots, n; n = 2t+1$, which implies $x_s^\top A_n x_s \geq x_s^\top A_s x_s$. This implies,

$$f(x_s) - f^* = f(x_s) - f_n^* \geq f_s^* - f_{2t+1}^* \geq f_t^* - f_{2t+1}^* \tag{1}$$

x^* is the unique solution in the span of $e_1, \dots, e_k$ of $A_k x = e_1$. It is easy to verify that it is defined by $x_k^*(i) = 1 - \frac{i}{k+1}$ for $i = 1, \dots, k$. Thus,

$$f_k^* = \frac{\beta}{8}(x_k^*)^\top A_k x_k^* - \frac{\beta}{4}(x_k^*)^\top e_1 = -\frac{\beta}{8}(x_k^*)^\top e_1 = -\frac{\beta}{8}\left(1 - \frac{1}{k+1}\right) \tag{2}$$

From definition of x_k^* for $k = n$, and $x_1 = 0$, we can derive lower bounds for $n \geq \frac{3}{2}\|x_1 - x_n^*\|_2^2$.
This bound, along with substitution of Equation 2 in Equation 1 resolves to Theorem (1).

Descent Lemma

The following lemma establishes the per-step bound for gradient descent on smooth functions. The original paper states the constrained case but for Nesterov's accelerated gradient descent in the smooth function case, we only need the unconstrained version.

Lemma (2)

Let $f: \mathbb{R}^n \to \mathbb{R}$ be a convex and β-smooth function. Let $x, y \in \mathbb{R}^n$ and $x^+ = x - \frac{1}{\beta}\nabla f(x)$. Then the following holds true:

$$f(x^+) - f(y) \leq \nabla f(x)^\top (x - y) - \frac{1}{2\beta}\|\nabla f(x)\|^2$$

Nesterov's accelerated gradient descent

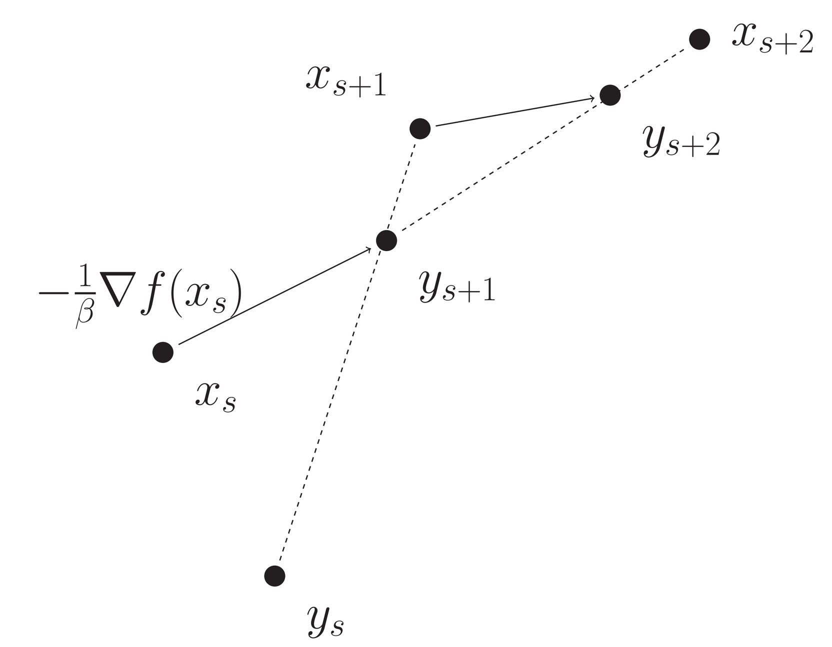

Figure: Illustration of Nesterov's accelerated gradient descent.

First we introduce the following sequences:

$$\lambda_0 = 0,\ \lambda_t = \frac{1 + \sqrt{1 + 4\lambda_{t-1}^2}}{2}, \text{ and } \gamma_t = \frac{1 - \lambda_t}{\lambda_{t+1}}$$

Nesterov's accelerated gradient descent is described as follows. Start at an arbitrary initial point $x_1 = y_1$ and then iterate the following equations for $t \geq 1$

$$y_{t+1} = x_t - \frac{1}{\beta}\nabla f(x_t) \tag{3}$$

$$x_{t+1} = (1 - \gamma_t) y_{t+1} + \gamma_t y_t \tag{4}$$

Nesterov's accelerated gradient descent uses the ideas from classical momentum gradient descent (Sutskever et al. 2013). x_{t+1} can be rewritten as $x_{t+1} = y_{t+1} - \gamma_t(y_{t+1} - y_t)$ which reads as x_{t+1} overshoots y_{t+1} along the vector $(y_{t+1} - y_t)$ with some "momentum" given by γ_t

Theorem (3)

Let f be a convex and β-smooth function, then Nesterov's accelerated gradient descent satisfies

$$f(y_t) - f(x^*) \leq \frac{2\beta\|x_1 - x^*\|^2}{t^2}$$

This gives us the $O(1/t^2)$ complexity. Write more about it here

Key insights from Proof of Theorem 3

- We can reverse engineer both λ_t and γ_t to show they are not arbitrary. Define $d_t := x_t - y_t$ (the momentum displacement) and $g_t := -\frac{1}{\beta}\nabla f(x_t)$ (the gradient step), so that $y_{t+1} = x_t + g_t$. Applying Lemma 2 with $x = x_t$ and $\nabla f(x_t) = -\beta g_t$ gives two inequalities:

$$f(y_{t+1}) - f(y_t) \leq -\frac{\beta}{2}\left(\|g_t\|^2 + 2\, g_t \cdot d_t\right) \tag{$\star$}$$

$$f(y_{t+1}) - f(x^*) \leq -\frac{\beta}{2}\left(\|g_t\|^2 + 2\, g_t \cdot (x_t - x^*)\right) \tag{$\star\star$}$$

- The big idea is to use the identity $\|a\|^2 + 2\,a \cdot b = \|a+b\|^2 - \|b\|^2$ on the combination of both inequalities to obtain a telescopic sum (Bubeck 2018). Let $\delta_t = f(y_t) - f(x^*)$. Multiplying ($\star$) by $(\lambda_t - 1)$ and adding ($\star\star$), and using $(\lambda_t - 1)\, d_t + x_t = y_t + \lambda_t d_t$, we obtain:

$$\lambda_t\, \delta_{t+1} - (\lambda_t - 1)\, \delta_t \leq -\frac{\beta}{2}\left(\lambda_t \|g_t\|^2 + 2\, g_t \cdot (y_t + \lambda_t d_t - x^*)\right)$$

  Multiplying both sides by λ_t and applying the identity with $a = \lambda_t g_t$, $b = y_t + \lambda_t d_t - x^*$:

$$\lambda_t^2\, \delta_{t+1} - (\lambda_t^2 - \lambda_t)\, \delta_t \leq -\frac{\beta}{2}\left(\|y_t + \lambda_t d_t + \lambda_t g_t - x^*\|^2 - \|y_t + \lambda_t d_t - x^*\|^2\right) \tag{$\dagger$}$$

- To obtain the telescopic sum, denote $v_t := \frac{\beta}{2}\|y_t + \lambda_t d_t - x^*\|^2$. We need:

$$y_t + \lambda_t(d_t + g_t) \ = \ y_{t+1} + \lambda_{t+1}\, d_{t+1}$$

  Since $y_{t+1} = y_t + d_t + g_t$ and $d_{t+1} = x_{t+1} - y_{t+1} = -\gamma_t(d_t + g_t)$:

$$\lambda_t(d_t + g_t) = (d_t + g_t) + \lambda_{t+1}\left[-\gamma_t(d_t + g_t)\right]$$

$$\lambda_t(d_t + g_t) = (d_t + g_t)(1 - \lambda_{t+1}\gamma_t)$$

  and hence $\gamma_t = \dfrac{1 - \lambda_t}{\lambda_{t+1}}$.

- With ($\dagger$) now telescoping as $\lambda_t^2\, \delta_{t+1} - (\lambda_t^2 - \lambda_t)\, \delta_t \leq v_t - v_{t+1}$, we require $\lambda_t^2 - \lambda_t = \lambda_{t-1}^2$ for the left side to collapse. Summing from $t = 1$ to $T-1$ and using $v_1 = \frac{\beta}{2}\|x_1 - x^*\|^2$ (since $d_1 = 0$) and $\lambda_{T-1} \geq T/2$ completes the proof.

References

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